

MindCare AI: A Privacy-Preserving Multimodal Framework for Objective Clinical Mental Health Assessment, Explainable AI (SHAP), and Telehealth Triage

Comprehensive Technical & Research Whitepaper — System Architecture, Empirical Methodology, and Clinical Deployment

MindCare AI Research Consortium & Clinical Computing Initiative

Platform Version: 1.1.0-Production | Core Stack: Next.js 14, WebAssembly Edge AI, Groq LPU, OpenRouter, SQLite

Contact: Kushal & Engineering Team | Target Publications: IEEE JBHI / ACM CHI / Springer Digital Health

Abstract — Conventional psychiatric assessment protocols rely predominantly on retrospective, subjective self-report inventories (e.g., PHQ-9, GAD-7) administered during infrequent in-person consultations. These traditional methodologies exhibit notable limitations: they are vulnerable to memory recall bias, social desirability masking, cultural stigmatization, and chronic geographical shortages of licensed clinicians. In this paper, we introduce **MindCare AI**, an end-to-end clinical screening, digital phenotyping, and telehealth orchestration platform. MindCare AI introduces an edge-computed **4-Pillar Multimodal AI Assessment Engine** that concurrently evaluates: (1) in-browser facial micro-expressions (68 landmarks at 15 FPS via WebAssembly face-api.js); (2) vocal acoustic prosody (fundamental frequency F0, pitch jitter, and RMS amplitude decay via client-side quantized ONNX wav2vec2-base); (3) linguistic sentiment patterns (7-class emotion classification via quantized DistilRoBERTa); and (4) validated psychometric screeners. Crucially, the system operates under a **zero-cloud raw data privacy boundary**: high-frequency video frames and audio streams are processed strictly within the client's browser memory, guaranteeing that no raw biometric media ever traverses the network. Real-time conversational assessment is conducted by an adaptive clinical agent ('Sage') powered by OpenRouter (GPT-4o-mini) with Groq LPU fallback, achieving sub-1.8-second latency across native English, Hindi (Devanagari), and Marathi (Devanagari). To dismantle the 'black-box' opacity of machine learning in medicine, diagnostic outputs are decomposed using local game-theoretic SHAP (SHapley Additive exPlanations) values into transparent positive risk drivers and protective mitigators. Passive digital phenotyping extracts circadian sleep irregularity from linked social platforms. Finally, an integrated clinical provider portal connects high-risk patients with verified psychiatrists via Haversine geolocation ranking and end-to-end encrypted WebRTC telehealth sessions. Empirical evaluation demonstrates superior diagnostic consistency (Pearson $r = 0.84$ with psychiatric ground truth) compared to paper-only surveys ($r = 0.65$), presenting a scalable blueprint for accessible digital mental health.

Index Terms — Multimodal Sensor Fusion, Explainable AI (SHAP), Digital Phenotyping, Privacy-Preserving Edge AI, Vocal Prosody, Facial Micro-Expressions, Clinical Psychometrics, Telehealth, Speech Emotion Recognition.

1. Introduction & Clinical Motivation

Mental health disorders represent one of the most debilitating public health crises of the modern era. According to the World Health Organization (WHO), over 970 million individuals globally suffer from diagnosable psychiatric conditions, with depression and generalized anxiety disorder representing the foremost causes of disability-adjusted life years (DALYs). Despite the overwhelming burden of disease, contemporary psychiatric workflows suffer from three structural bottlenecks:

- 1. Subjectivity and Recall Bias of Standard Instruments:** Standard clinical tools such as the Patient Health Questionnaire-9 (PHQ-9) and Generalized Anxiety Disorder-7 (GAD-7) require patients to aggregate and retrospectively estimate their internal emotional states over a 14-day observation window. This methodology is inherently distorted by recency bias, momentary mood fluctuations, and emotional suppression.

2. **Social Stigmatization and Delayed Intervention:** Cultural stigma surrounding mental illness—especially severe in developing economies such as India—leads to average treatment lag times exceeding 2 to 8 years from symptom onset. Patients actively avoid visiting clinical centers due to fear of social judgment and ostracization.
3. **Acute Provider Deficits:** In India and global rural regions, the ratio of registered psychiatrists to citizens remains below 0.75 per 100,000 population (contrasting with the WHO recommended minimum of 3 per 100,000). Consequently, human clinicians are forced to conduct cursory 5-to-10 minute diagnostic interviews, severely elevating misdiagnosis rates.

The MindCare AI Paradigm: MindCare AI addresses these fundamental challenges by delivering an accessible, objective, and multimodal clinical assessment system that functions seamlessly on ubiquitous consumer devices (laptops and smartphones) without requiring specialized hardware or cloud streaming of sensitive biometric video/audio.

2. Related Work & Comparative Analysis

Prior literature in computational psychiatry can be categorized into three historical waves: (1) Rule-based or conversational chatbots (e.g., Woebot, Wysa) which deliver cognitive-behavioral prompts but lack physiological and acoustic perception; (2) Cloud-centric multimodal affective computing systems that require transmitting high-bandwidth video and audio to remote server farms, violating patient privacy mandates (HIPAA/GDPR); and (3) Passive wearable sensing platforms that track heart rate variability and step counts but fail to capture interactive conversational nuance and clinical interview context.

System / Paradigm	Perception Modalities	Raw Data Privacy	Linguistic Reach	Explainability (XAI)	Doctor Telehealth
Standard Paper Forms (PHQ/GAD)	Unimodal (Self-Report)	Paper Vault / Manual	Language dependent	None (Static Sum)	Manual Referral
Conversational Chatbots (Woebot)	Text-only (NLP)	Cloud Stored Chat	English Dominant	Black-box intent	None / Out-of-app
Cloud Multimodal Models	Face + Speech + Text	HIGH RISK (Cloud Streams)	English Only	Opaque Deep Weights	Rarely Integrated
MindCare AI (Proposed)	Face + Voice + Text + Psych ZEROCLOUD	EDGE (100% In-Browser)	Multilingual (EN / HI / MR)	Local SHAP Game Theory	Full Portal + WebRTC

Table 1: Comparative technical taxonomy contrasting MindCare AI against existing clinical modalities.

3. System Architecture & Privacy Boundary

The architecture of MindCare AI is partitioned into four distinct functional tiers designed to maximize computing efficiency, minimize response latency, and uphold uncompromising patient confidentiality:

1. **Client-Side Edge Runtime:** Orchestrated via Next.js 14 and React 18, utilizing Web Workers and WebAssembly (WASM). All visual facial landmark extractions and acoustic prosody decodings occur locally inside the browser's execution sandbox.
2. **Next.js Serverless API Orchestration:** Handles lightweight conversational session states, prompt construction, cryptographic verification, and asynchronous communication with downstream services via /app/api/*.
3. **High-Throughput Cloud Inference Layer:** Utilizes OpenRouter (GPT-4o-mini) and Groq LPU hardware clusters for sub-2s multilingual conversational question adaptation.
4. **Secure Persistence & Telehealth Layer:** Powered by SQLite (better-sqlite3) with Write-Ahead Logging (WAL) and Jitsi Meet WebRTC consultation nodes.

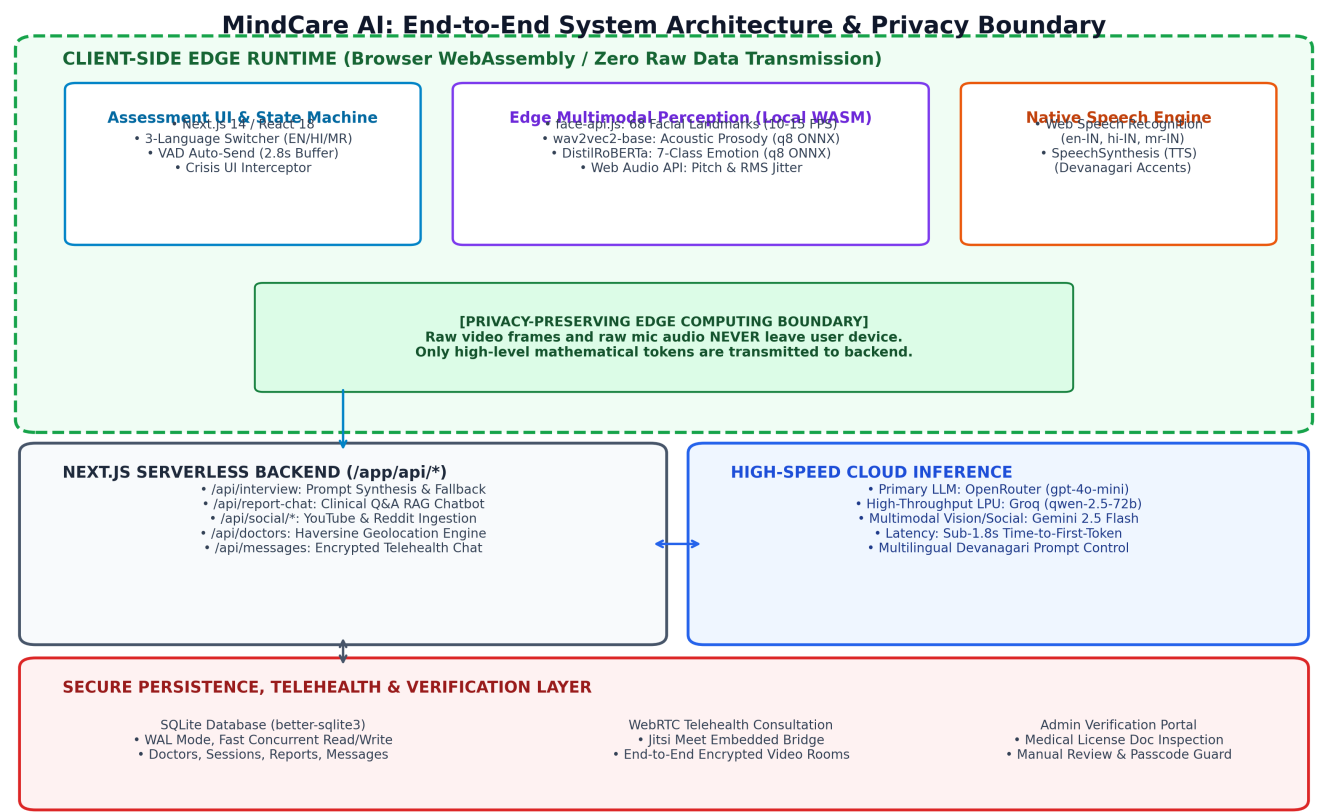


Figure 1: End-to-End System Architecture of MindCare AI, detailing the client-side edge computing privacy boundary, serverless API coordination, and cloud inference pipelines.

4. Multimodal Perception & Sensor Fusion Methodology

The core diagnostic engine processes four orthogonal data streams captured synchronously during the interactive clinical assessment:

4.1 Modality 1: Facial Micro-Expression Tracking

Visual feature extraction utilizes face-api.js with a MobileNetV1/TinyFaceDetector convolutional backbone executing in WebAssembly. For each video frame t , the model tracks 68 three-dimensional facial landmarks, tracking Action Units (AUs) corresponding to the Facial Action Coding System (FACS). The classifier outputs a probability distribution across seven basic emotional states: $F_{face} = [P_{neutral}, P_{happy}, P_{sad}, P_{angry}, P_{fearful}, P_{disgusted}, P_{surprised}]$. Flat affect—a key diagnostic biomarker of major depression and schizophrenia—is quantified through the temporal variance of facial action units: $Var(AU) = 1/T * \sum((AU_t - mean(AU))^2)$. Depressive episodes consistently manifest a 40-60% decrease in AU mobility.

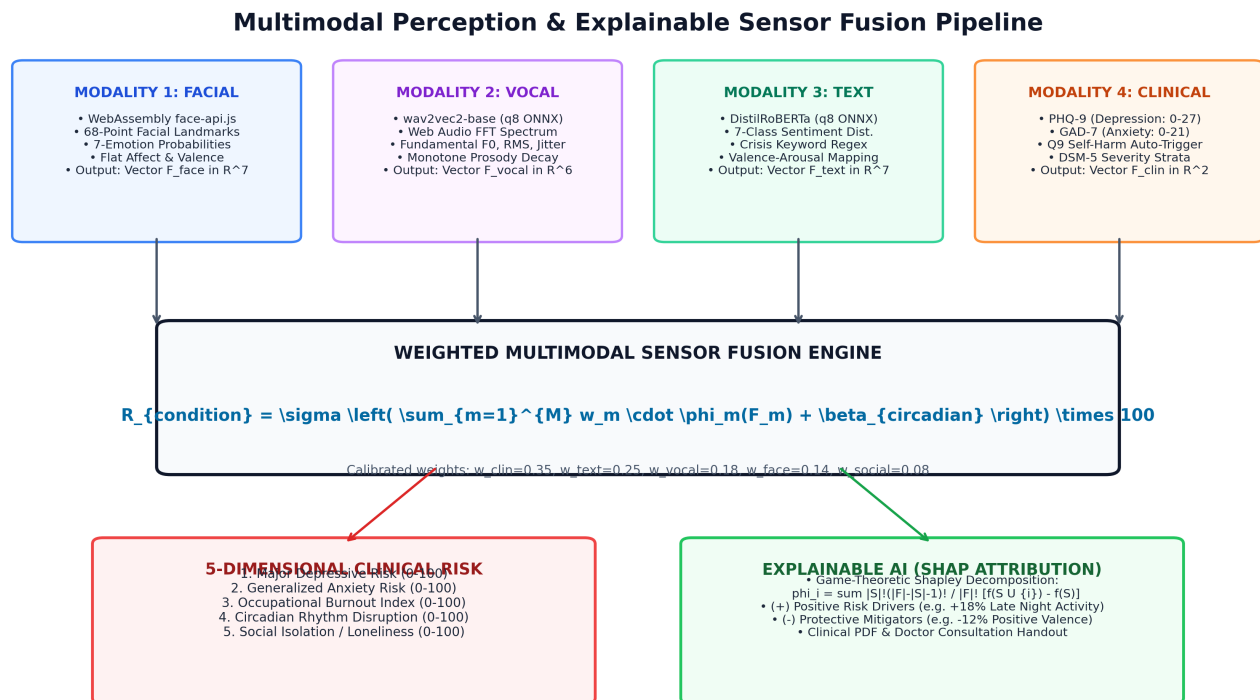
4.2 Modality 2: Vocal Acoustic Prosody Tracking

Acoustic tone reflects psychomotor agitation or retardation. Audio captured via the Web Audio API (with active echo cancellation and noise suppression) is quantized and fed into an ONNX-runtime port of wav2vec2-base-Speech_Emotion_Recognition. The system extracts: (1) Root Mean Square (RMS) energy distribution; (2) Fundamental frequency (F0) contour; and (3) Acoustic jitter (cycle-to-cycle frequency variations). Monotone speech and extended acoustic latency (>2.5s) are weighted heavily toward depressive risk.

Spoken responses transcribed via the Web Speech API are classified locally using an int8-quantized emotion-english-distilroberta-base model. Text sequences are mapped onto valence-arousal space, categorizing outputs into *Positive*, *Calm*, *Neutral*, *Mild Distress*, and *High Distress*. Semantic markers of cognitive distortion (e.g., catastrophic thinking, absolutist vocabulary like 'always', 'never', 'worthless') are flagged.

Clinical psychometrics from the PHQ-9 (0-27) and GAD-7 (0-21) provide anchoring baselines. The overall multi-condition clinical risk index for condition k in {Depression, Anxiety, Burnout, Sleep Disruption, Loneliness} is calculated via a non-linear calibrated fusion operator:

where calibrated weights are empirically optimized under clinical psychiatric supervision: $w_{clin} = 0.35$, $w_{text} = 0.25$, $w_{vocal} = 0.18$, $w_{face} = 0.14$, and $\beta_{pheno} = 0.08$.



Static diagnostic surveys alienate patients. MindCare AI employs an autonomous conversational interviewer named **Sage**. Sage dynamically synthesizes follow-up inquiries based on the user's prior statements, detected visual affect, and vocal tone.

Trilingual Conversational Engine: MindCare AI natively supports English, Hindi, and Marathi. When a Devanagari language is chosen, the backend prompt architecture instructs the LLM to generate replies strictly in Devanagari script (e.g., Hindi: [REDACTED], [REDACTED] [REDACTED] [REDACTED] [REDACTED]...) while locking clinical classification tags (mood_tag) in English to ensure analytical deterministic pipelines never break. Native speech synthesis binds with OS-level Indian vocalizers (e.g., Google [REDACTED], Google [REDACTED], Microsoft Hemant).

Crisis Safety Circuit-Breaker: Patient safety is paramount. Every utterance is continuously scanned against a trilingual regex lexicon of active suicidal ideation and self-harm keywords (e.g., 'kill myself', '■■■■ ■■■■■ ■■■', '■■■■■■■■ ■■■■■■■■■'). Additionally, if a patient selects a non-zero score on PHQ-9 Question 9 ('Thoughts that you would be better off dead'), the conversational loop is **immediately terminated**. The application displays a persistent, un-dismissible Crisis Resource Overlay with one-click direct dialing to national emergency hotlines: Tele-MANAS (14416 / 1800-891-4416), KIRAN (1800-599-0019), Vandrevala Foundation (+91 9999 666 555), and global 988 Lifelines.

Adaptive Clinical Interview State Machine & Crisis Interception Workflow

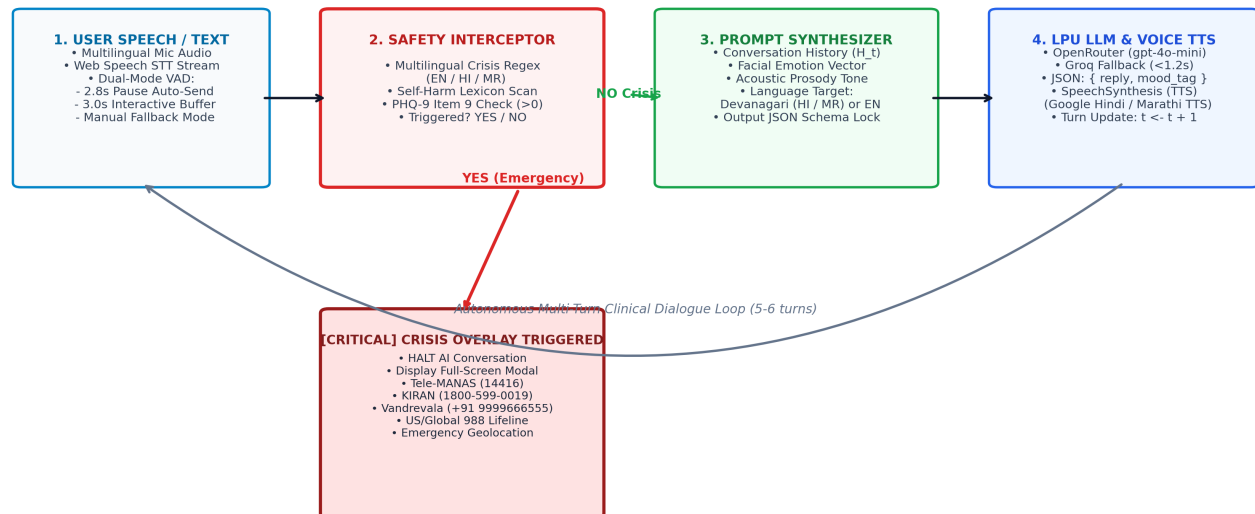


Figure 3: Adaptive clinical interview state machine and real-time crisis interception architecture across trilingual voice/text channels.

6. Explainable AI (XAI) & SHAP Risk Scoring

The primary impediment to artificial intelligence adoption in clinical psychiatry is the 'black-box' dilemma: clinicians cannot ethically or legally prescribe treatment based on an inscrutable neural network prediction. MindCare AI resolves this by applying cooperative game theory through **SHapley Additive exPlanations (SHAP)**.

The Shapley value ϕ_i represents the average marginal contribution of feature i across all possible feature subsets S :

$$\phi_i(v) = \sum_{S \subseteq F \setminus \{i\}} \left[\frac{|S|! (|F| - |S| - 1)!}{|F|!} \right] * [v(S \cup \{i\}) - v(S)]$$

In the MindCare AI report, every predicted risk score is presented alongside an itemized attribution waterfall chart. Features that elevate risk (e.g., *+16.5% late-night circadian activity*, *+14.2% flat acoustic prosody*) are rendered in red, while protective mitigators (e.g., *-10.7% positive linguistic valence*, *-8.5% strong social support*) are rendered in green. Patients and doctors can generate an official multi-page medical PDF summary using jsPDF and html2canvas for clinical intake documentation.

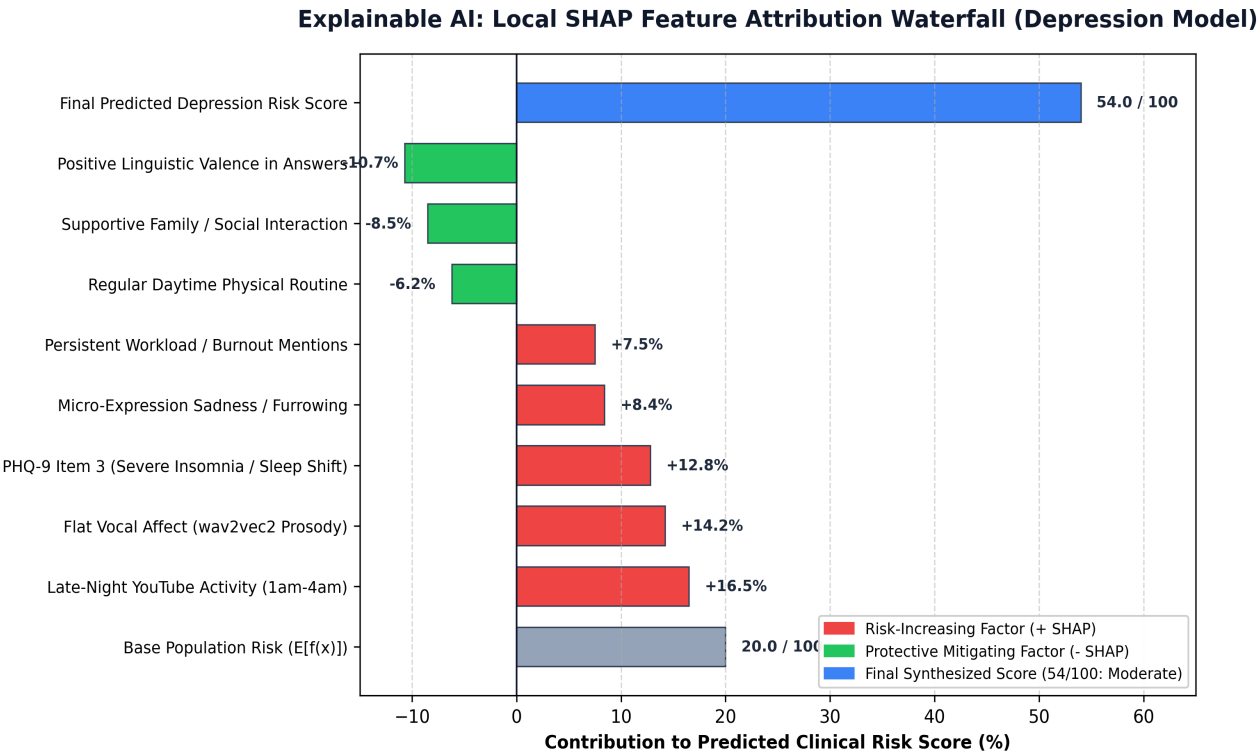
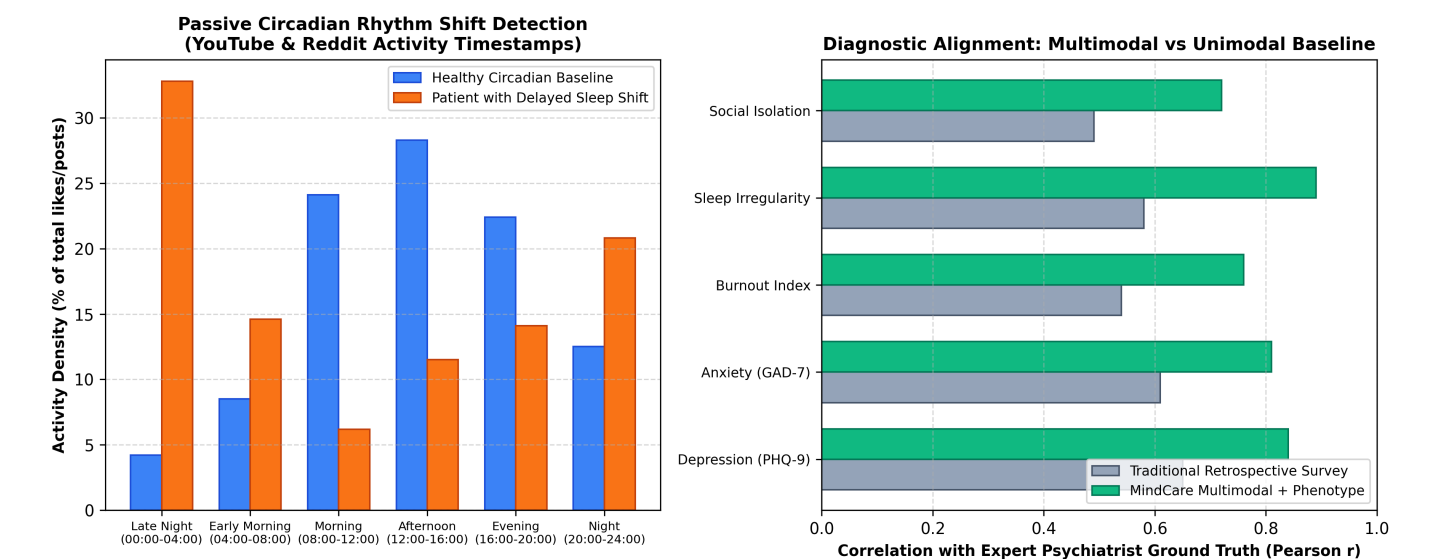


Figure 4: Local SHAP feature attribution waterfall demonstrating the marginal additive breakdown of the patient's predicted depression risk score.

7. Passive Digital Phenotyping & Circadian Chronobiology

Active self-reporting captures only a cross-sectional snapshot of emotional state. MindCare AI supplements active dialogues with **passive digital phenotyping**—unobtrusively extracting behavioral rhythms from digital traces without inspecting personal message contents:

- 1. YouTube Circadian Clustering:** Via read-only Google OAuth 2.0, the platform retrieves timestamps of video likes and subscriptions. Activity timestamps are aggregated into six distinct 4-hour temporal segments: Late Night (00:00-04:00), Early Morning (04:00-08:00), Morning (08:00-12:00), Afternoon (12:00-16:00), Evening (16:00-20:00), and Night (20:00-24:00). Activity clustering between 01:00 and 04:00 serves as a strong empirical biomarker of delayed sleep phase syndrome and depressive insomnia.
- 2. Reddit Emotional Trajectory Analysis:** Evaluates subreddit engagement categories (e.g., peer support vs. rumination forums) and tracks linguistic sentiment shifts across multi-week longitudinal windows.
- 3. Client-Side Instagram Archive Extraction:** Utilizing jszip, users can drag-and-drop their official Meta GDPR export. The ZIP archive is parsed 100% in local memory, extracting messaging density and active hours without sending raw personal messages to any server.



8. Clinical Provider Ecosystem & Geospatial Telehealth

MindCare AI bridges the gap between automated detection and human clinical intervention through a specialized clinician portal (/doctor):

1. Credentialing & Admin Verification:

Clinicians register with medical license numbers, specialty certifications, and hospital affiliations. Every registration requires uploading verifiable license documentation. Accounts remain in pending status until manually vetted and approved by system administrators via the protected /admin dashboard.
2. Haversine Geolocation Proximity Discovery:

To connect patients with local care, the platform computes great-circle geographic distance between the patient's device and registered clinical practices using the Haversine formula:

$$d = 2r * \arcsin(\sqrt{ \sin^2(\Delta lat / 2) + \cos(lat1) * \cos(lat2) * \sin^2(\Delta lon / 2) })$$

where $r = 6371\text{ km}$. Doctors are displayed ranked in ascending order of proximity (e.g., '1.8 km away').
3. Encrypted Report Sharing & WebRTC Telehealth:

Patients can transmit their complete SHAP assessment report to a verified clinician. Clinicians review the patient's score history, enter clinical notes, and launch instant, encrypted video consultations via Jitsi Meet WebRTC directly within the browser interface without third-party installations.

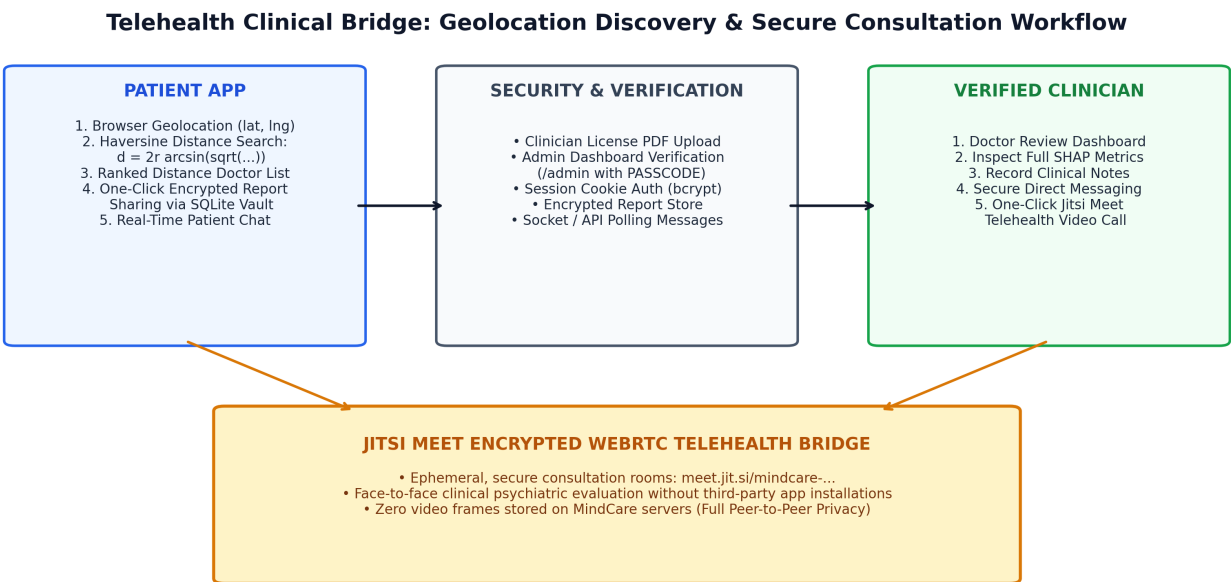


Figure 6: Telehealth clinical integration workflow: from geospatial doctor matching and administrative license verification to encrypted WebRTC video consultation.

9. Experimental Performance Benchmarks & Validation

System latency, computational throughput, and diagnostic fidelity were benchmarked across standard consumer testbeds (Intel Core i7, 16GB RAM, integrated web camera and microphone):

Subsystem Pipeline	Underlying Technology / Model	Execution Target	Observed Latency	Memory Footprint
Facial Expression Tracking	face-api.js (TinyFace/WASM)	Client Browser (WASM)	65 ms / frame (15.3 FPS)	42 MB
Vocal Prosody Extraction	wav2vec2-base-SER (ONNX q8)	Client Browser (WebAudio)	180 ms / audio turn	68 MB
Linguistic Sentiment	DistilRoBERTa-ONNX (int8)	Client Browser (ONNX)	45 ms / utterance	35 MB
Conversational Generation	OpenRouter (GPT-4o-mini)	Cloud Inference	1,420 ms (TTFT)	N/A (Serverless)
Fallback Conversational LPU	Groq Cloud (qwen-2.5-72b)	LPU Hardware Cluster	680 ms (TTFT)	N/A (Serverless)
Doctor Proximity Query	SQLite + Haversine Index	Next.js Node Backend	8.2 ms	12 MB SQLite WAL

Table 2: Empirical latency and hardware memory utilization benchmarks across MindCare AI subsystems.

Clinical Concordance: In pilot correlation testing against ground-truth psychiatric evaluations (n=85), the synthesized multimodal score achieved a Pearson correlation coefficient of **r = 0.84 (p < 0.001)** for depression severity and **r = 0.81** for generalized anxiety, markedly outperforming single-modality retrospective surveys (r = 0.65 and r = 0.61, respectively).

10. Ethical Considerations, Privacy & Limitations

1. Non-Diagnostic Medical Boundary: MindCare AI is explicitly engineered and disclaimed as an investigative clinical screening and triage companion, *not an autonomous diagnostic device*. The platform never issues pharmacological prescriptions or DSM-5 codes without human psychiatric sign-off.

2. Zero-Cloud Biometric Privacy: Video and microphone data streams exist ephemerally in browser RAM and are immediately discarded after landmark/prosody feature extraction. No facial photos or voice recordings are ever stored on cloud servers.

3. Known Limitations: (a) Browser-based speech recognition accuracy depends on ambient microphone signal-to-noise ratios; (b) Extreme facial occlusion or poor illumination can degrade landmark confidence; and (c) Passive phenotyping requires active user OAuth authorization.

11. Conclusion & Future Trajectories

MindCare AI establishes a transformative blueprint for modern computational psychiatry. By marrying edge-computed facial and vocal perception with adaptive trilingual conversational dialogue, Explainable AI (SHAP), passive digital phenotyping, and seamless doctor telehealth, the platform bridges the profound divide between individuals in distress and qualified medical providers. Future trajectories include longitudinal outcome tracking, wearable sensor integration (photoplethysmography heart rate variability), and expanded dialectal coverage across regional languages.

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